Chandra Sekhar Bhamidi

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Physical Design and STA Engineer with 2+ years of hands-on experience in advanced-node ASIC implementation and timing closure, specializing in 7nm FinFET CPU physical design and RTL-to-GDSII implementation. Executed the complete physical design flow for an ARM-based CPU core with 3.5M+ instances, 350+ macros, and 13 metal layers, achieving 2.54 GHz against a 2.60 GHz target. Expertise across floor planning, placement optimization, routing, MCMM timing analysis, timing closure, post-route ECOs, parasitic extraction, and signoff, using Synopsys Fusion Compiler, IC Compiler II, PrimeTime, StarRC, Design Compiler, and Formality. Strong scripting experience with TCL for timing-report parsing, path-group analysis, and QoR automation.

Physical Design & Implementation: RTL-to-GDSII | Floorplanning | Macro Placement | Placement Optimization | Routing | Route Optimization | QoR Optimization | Physical Implementation | Post-Route ECO

STA / Timing Closure: PrimeTime | MCMM Analysis | Setup Timing | WNS / TNS Analysis | Path-Group Analysis | Clock Uncertainty | Timing Constraints | Timing ECO | Cell Upsizing | VT Swaps | Buffer Insertion | Timing Signoff

Synopsys EDA: Fusion Compiler | IC Compiler II | Design Compiler | PrimeTime | StarRC | Formality

Automation & Scripting: TCL | Timing Report Parsing | Path-Group Filtering | QoR Automation | Flow Automation

Technology / Design: 7nm FinFET | 28nm | 13 Metal Layers | Multi-Corner Multi-Mode | PVT Analysis | SPEF | RC Parasitic Extraction | DFT Timing

Physical Verification: DRC Awareness | Post-Route Analysis | ECO Implementation | Physical Design Signoff

EXPERIENCE HIGHLIGHTS

Synopsys Tokyo, Tokyo, Japan | November 2024 – Present

Applications Engineer (Project: ARM-based CPU Core)

Technology & Tools: 7nm FinFET | Fusion Compiler | IC Compiler II | PrimeTime | StarRC

- Executed the complete RTL-to-GDSII physical design flow for an ARM-based CPU core with 3.5M+ instances, 350+ macros, and 13 metal layers.
- Achieved 2.54 GHz without DFT scan and 2.50 GHz with DFT scan against a 2.60 GHz target, closing timing across 4+ MCMM PVT corners.
- Performed 20+ floorplan iterations across island-style, edge-based, hybrid, and freestyle approaches to optimize macro locality and critical-path wire length.
- Developed TCL automation to filter and rank worst-violating path groups by WNS, improving timing-debug turnaround.
- Analyzed WNS/TNS sensitivity by sweeping clock uncertainty across the physical design flow to identify closure margins across 4+ PVT corners.
- Applied incremental placement optimization during initial and final optimization stages and performed multiple route optimization iterations to improve timing convergence.
- Executed post-route timing ECOs, including cell upsizing, VT swaps, and buffer insertion, targeting the top critical setup paths.
- Applied path-group optimization weighting to prioritize timing closure on the most critical violating groups during signoff.
- Debugged DFT-induced timing degradation and applied appropriate false-path constraints on scan ports to recover timing budget.
- Generated post-route SPEF using Synopsys StarRC and integrated extracted RC parasitics into PrimeTime for multi-corner timing signoff.

ZOICX Technologies, Hyderabad, India | June 2024 – October 2024

Digital Implementation Trainee (Project: ARM Core)

Technology & Tools: 28nm | Design Compiler | IC Compiler II | Formality

- Executed the physical design flow for a 300K-gate ARM core with 29 macros and 9 metal layers, achieving the 500 MHz target.
- Performed logic synthesis using Design Compiler and verified post-synthesis netlist equivalence using Synopsys Formality.
- Performed floorplan optimization through 10+ trials across core shapes and utilization targets to address routing congestion.
- Benchmarked OpenROAD, Yosys, and OpenSTA against Synopsys commercial tools across synthesis, physical implementation, and STA stages.

RiseTime Semiconductor | Hyderabad, India | Oct 2023 - Mar 2024

Physical Design Engineer – Trainee

Technology & Tools: 28nm | 9 Metal Layers | Design Compiler | IC Compiler II

- Executed synthesis-to-post-route physical design flow for a 72K-gate design with 6 macros and 2 clocks at 454 MHz, covering synthesis, floorplanning, placement, CTS, routing, and timing checks.
- Resolved missing-via, pin-density, and congestion issues through keepout-margin optimization and multiple floorplan trials with different core shapes and utilization targets.

Key Project:

ARM-Based CPU Core — Full Physical Design | 7nm FinFET | Fusion Compiler | IC Compiler II | PrimeTime | StarRC

- Executed RTL-to-GDSII implementation for a 3.5M+ instance / 350+ macro CPU design, including floor planning, placement, routing, MCMM timing closure, RC extraction, and post-route ECO implementation.
- Achieved 2.54 GHz without DFT and 2.50 GHz with DFT against a 2.60 GHz target across 4+ PVT corners; achieved clean post-routing DRC and completed manual timing ECOs.

Design Metrics:

3.5M+ Instances | 350+ Macros | 7nm FinFET | 13 Metal Layers | Flat Design | 2–4 Engineer Team | MCMM 4+ PVT Corners

EDUCATION & CREDENTIALS

B. Tech — Electronics & Communication Engineering: CVR College of Engineering, Hyderabad (Aug 2022)

Post Graduate Diploma, Physical Design, Sumedha Institute of Technology (Oct 2022 – May 2023)

NPTEL Certified: VLSI Design Flow RTL to GDS (66%), *November 2023*

Publication: IEEE — J-Pole Antenna for ADS-B Signals, *April 2023*